

Time: 3 HOURS

Max marks: 60

Note: Solve any two parts from each question. All questions carry equal marks. Make suitable assumptions whenever required. Internal choice is given in each question.

- Q.1 (a) Explain NOR and NAND logic with the help of logic symbol and truth table and prove its Universality of the gates. 6
- Q.1(b) Explain ASCII code and Gray codes at least one suitable examples. 6
- Q.1(C) Solve .. 6
- (i) Convert Decimal number  $(529.29)_{10}$  into equivalent binary number.
  - (ii) Convert number  $(564.37)_8$  into equivalent Hexadecimal number.
  - (iii) convert number  $(9623.636)_{10}$  into equivalent octal number.
  - (iv) convert number  $(45A37.C3)_{16}$  into equivalent Decimal number.
  - (v) subtract 111011 from 101010 using 1's complement.
  - (vi) Add (+125) and (-68) using 2's complements.
- Q.2 (a) Design full subtractor circuit using two half subtracters circuits and an OR gate and explain its working with two different input combinations. 6
- Q.2(b) Give simplified expression of  $Y = F(A, B, C, D) = \sum m(1, 2, 8, 9, 10, 12, 13, 14)$  using Quine-McClusky method. 6
- Q.2(C) Design 2bit  $A = A_1A_0$  and  $B = B_1B_0$ , Magnitude comparator using truth table and circuit diagram. 6
- Q.3 (a) What is the primary difference between a JK and an RS flip-flop? Explain working of positive edge triggered JK flip flop with the help of truth table and circuit diagram. 4
- Q.3 (b) Design a 3-bit binary ripple down counter using T flip flop that trigger on the Positive edge. 6
- Q.3(C) Design modulus-8 binary synchronous Down counter using T flip flop. 6
- Q.4 (a) Design bi-Directional shift register and implement a 4-bit bidirectional shift register and perform the following operations with the help of signal diagram: 6
- 1. Load initial data of 1101.
  - 2. Shift the data to the right twice.
- Q.4(b) Design parallel in serial out shift register and explain its working with the help of timing diagram. 6
- Q.4(C) What is the difference between serial and parallel data transfer? What type of register is used in each case? 6
- Q.5 Write short note on any two. 6
- (a) 4 - bit Binary Weighted Digital to analogue converter 6
  - (b) Flash ADC 6
  - (c) Successive Approximation type ADC 6